

ESTIMATION OF THE SELECTED OF TWO NORMAL MEANS

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Summary: Suppose independent samples from two normal populations are available and the true mean of the population whose sample yielded the largest sample mean must be estimated. For the case when the sample sizes and the population variances are equal we introduce a limited translation estimator which is based on the work of Bickel (1984). This estimator is approximately optimal in the sense of minimizing the mean squared error when the true population means are equal subject to having its global mean squared error not exceeding the minimax value by more than a prescribed amount. In this sense it outperforms all existing estimators. The results are generalized to the cases when the sample sizes and population variances are not necessarily equal and may be unknown. These cases have not been treated in the literature before.

1. INTRODUCTION

Suppose we have two independent samples of sizes n_1 and n_2 from $N(\theta_1, \sigma_1^2)$ - and $N(\theta_2, \sigma_2^2)$ -distributed populations respectively. Let X_1 and X_2 be the two sample means and S_1^2 and S_2^2 the two unbiased sample variances. Also let ℓ denote the index of the largest sample mean, i.e.

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$$(1.1) \quad X_\ell = \max(X_1, X_2).$$

Then population ℓ is called the "selected population". The importance of the problem of estimating θ_ℓ , the true mean of the selected population, is discussed in Cohen and Sackrowitz (1982) where further references to the existing literature can be found. See also Sackrowitz and Samuel-Cahn (1986) for related work in this context.

With squared error loss the problem of estimating θ_ℓ is equivariant under simultaneous translation of both X_1 and X_2 . It therefore seems reasonable to restrict attention to translation equivariant estimators and such estimators must be of the form

$$(1.2) \quad X_\ell - h_\ell(|X_1 - X_2|)$$

with h_1 and h_2 suitable functions. A naive estimator of θ_ℓ is X_ℓ itself. However, $EX_\ell \geq EX_\lambda = \theta_\lambda \geq E\theta_\ell$ where $\theta_\lambda = \max(\theta_1, \theta_2)$. So X_ℓ is biased upwards and tends to overestimate θ_ℓ . We may think of $h_\ell(|X_1 - X_2|)$ in (1.2) as an adjustment to correct for this tendency to overestimate and we call h_ℓ the "adjustment function" of the estimator (1.2).

For the case when $\sigma_1^2/n_1 = \sigma_2^2/n_2$ and this common value is known, a number of estimators have been proposed (e.g. Dahiya (1974), Cohen and Sackrowitz (1982) and Venter (1988)). In §2 we list the adjustment functions of these estimators. An experimenter who believes that the true means do not differ markedly, would prefer to use an estimator that has small mean squared error (MSE) when $\theta_1 = \theta_2$. In such circumstances the various estimators of the selected mean that are available, should be judged by fixing their maximal MSE's at a prescribed level and comparing their MSE's when $\theta_1 = \theta_2$. We show that the problem of choosing an optimal adjustment function from this point of view reduces to Problem P of Bickel (1984) and derive a nearly optimal choice based on the limited translation function of Efron and Morris (1971). A comparison of this new estimator with its competitors shows that it outperforms them.

In §3 we consider the case where it is not assumed that $\sigma_1^2/n_1 = \sigma_2^2/n_2$ but these quantities are still assumed known. This case has not been treated

in the literature before. We show that X_L is still an admissible minimax estimator and introduce a limited translation type estimator for this case as well. In §4 we extend the limited translation estimator to the case where $\sigma_1^2 = \sigma_2^2$ but they are otherwise unknown. Finally §5 considers the case when this assumption is also dropped.

The use of ℓ to select a population in the heteroscedastic case may be questioned in the sense that it is not an optimal selection rule under these circumstances. Cohen and Sackrowitz (1988) and Gupta and Miescke (1988) discuss some of the problems surrounding the use of this so-called "natural selection rule". Gupta and Miescke (1988) state, however, that "... no modification or substitute of the natural selection rule has been found so far which can be considered to be better in some reasonable sense." We will therefore limit our discussion in this paper to the case where ℓ is used as selection rule.

2. THE CASE WHERE $\sigma_1^2/n_1 = \sigma_2^2/n_2$ AND THESE ARE KNOWN

In this paragraph we assume that $\sigma_1^2/n_1 = \sigma_2^2/n_2 = b^2$ and that b is known. The problem of estimating θ_L is then also invariant under permutation of X_1 and X_2 and we further restrict attention to estimators symmetric in X_1 and X_2 . This implies that h_L does not depend on ℓ and the estimators have the simpler form $X_L - h(|X_1 - X_2|)$. Dahiya (1974) discusses four estimators corresponding to the following adjustment functions:

$$h^{(1)}(y) = 2bc\phi(y/2b)$$

$$h^{(2)}(y) = \sqrt{2}bc\phi(cy/\sqrt{2}b) + y\Phi(-cy/\sqrt{2}b)$$

$$h^{(3)}(y) = y\Phi(-y/\sqrt{2}b) + c[b\phi(y/2b) + y\{\Phi(y/2b) - \Phi(y/\sqrt{2}b)\}]$$

$$h^{(4)}(y) = \frac{1}{2}yI(y \leq \sqrt{2}bc)$$

where $c > 0$ is a constant and ϕ and Φ are the $N(0,1)$ -density and distribution functions respectively while I is the indicator function. Venter (1988) studies an estimator which corresponds to

$$h^{(5)}(y) = \sqrt{2}bc\phi(cy/\sqrt{2}b)$$

and Cohen and Sackrowitz (1982) introduce a class of estimators with

adjustment functions of the form

$$h^{(6)}(y) = y/(2 + r(y/b))$$

with $r(y)$ a non-decreasing function for $y \geq 0$ with $r(0) = 0$ and $r(y) \rightarrow \infty$ as $y \rightarrow \infty$. Specific choices mentioned are $r(y) = cy^d$ with $c > 0$ and $d \geq 1$ and $r(y) = (y^2 - 2)I(y \geq 2)$.

How can we choose from this plethora of possibilities? It is known (Cohen and Sackrowitz (1982), p.244) that X_L is admissible and minimax with constant MSE b^2 . It is convenient to express all MSE's relative to this minimax value b^2 . So the relative MSE (RMSE) of an estimator is its MSE divided by b^2 . Then X_L has the constant RMSE 1 and no estimator exists with a RMSE uniformly below 1. The experimenter who believes that the true treatment means do not differ markedly, attaches particular importance to the configuration $\theta_1 = \theta_2$. Such an experimenter will therefore act in the spirit of Bickel (1984) and set himself the objective of finding an estimator that is optimal in the sense that it has minimal MSE on the line $\theta_1 = \theta_2$ subject to its maximal RMSE not rising to a level above $1 + \epsilon$, for some $\epsilon > 0$. The estimators corresponding to $h^{(j)}$ above can all be judged from this point of view, i.e. the constants c may be chosen in such a way that their maximal RMSE's are all $1 + \epsilon$ and then their RMSE's on the line $\theta_1 = \theta_2$ can be investigated. In fact we can pose the question: which adjustment function is optimal in the sense that among all choices for which the RMSE of the resulting estimator is at most $1 + \epsilon$, the RMSE when $\theta_1 = \theta_2$ is minimal? This problem is equivalent to Problem P of Bickel (1984). To see this, put

$$(2.1) \quad Y = (X_1 - X_2)/\sqrt{2}b \text{ and } \Delta = (\theta_1 - \theta_2)/\sqrt{2}b$$

so that Y is $N(\Delta, 1)$ -distributed. Then the RMSE of $X_L - h(|X_1 - X_2|) = X_L - h(\sqrt{2}b|Y|)$ is

$$\begin{aligned} (2.2) \quad R(h, \Delta) &= E[X_L - \theta_L - h(\sqrt{2}b|Y|)]^2/b^2 \\ &= \{E[X_1 - \theta_1 - h(\sqrt{2}bY)]^2 I(Y \geq 0) \\ &\quad + E[X_2 - \theta_2 - h(-\sqrt{2}bY)]^2 I(Y < 0)\}/b^2 \end{aligned}$$

Taking conditional expectations given Y , the RMSE in (2.2) becomes

$$(2.3) \quad \frac{1}{2} \{ 1 + E[Y - \Delta - (\sqrt{2}/b)h(\sqrt{2}bY)]^2 I(Y \geq 0) \\ + E[Y - \Delta + (\sqrt{2}/b)h(-\sqrt{2}bY)]^2 I(Y < 0) \}.$$

Introduce the "standardized adjustment function" $\tilde{h}$ related to h by

$$(2.4) \quad \tilde{h}(y) = (\sqrt{2}/b) \{ h(\sqrt{2}by)I(y \geq 0) - h(-\sqrt{2}by)I(y < 0) \}.$$

Then the RMSE of $X_t - h(|X_1 - X_2|)$ becomes

$$(2.5) \quad R(h, \Delta) = \frac{1}{2} \{ 1 + E[Y - \Delta - \tilde{h}(Y)]^2 \}.$$

This is at most $1 + \epsilon$ if and only if

$$(2.6) \quad E[Y - \Delta - \tilde{h}(Y)]^2 \leq 1 + 2\epsilon.$$

Since Y is $N(\Delta, 1)$ -distributed, (2.5) depends only on Δ . Choosing $\tilde{h}$ to minimize the left hand side of (2.6) at $\Delta = 0$ (i.e. $\theta_1 = \theta_2$) subject to (2.6) holding for all Δ , is exactly Bickel's Problem P with $q = \sqrt{2\epsilon}$. Bickel (1984) points out that the exactly optimal solution to this problem is complicated but argues from various points of view that an approximately optimal solution is provided by the Efron-Morris limited translation function see (Efron and Morris (1971))

$$(2.7) \quad \tilde{h}(y) = yI(|y| \leq \sqrt{2\epsilon}) + \sqrt{2\epsilon}I(y > \sqrt{2\epsilon}) - \sqrt{2\epsilon}I(y < -\sqrt{2\epsilon}).$$

The related adjustment function h follows from (2.4) as

$$(2.8) \quad h^{(7)}(y) = \frac{1}{2}yI(0 \leq y \leq 2b\sqrt{\epsilon}) + b\sqrt{\epsilon}I(y > 2b\sqrt{\epsilon}).$$

The estimator corresponding to this adjustment function will be called the limited translation (LT) estimator. Note that if we take $r(y) = (y/\sqrt{\epsilon} - 2)I(y > 2\sqrt{\epsilon})$ in $h^{(6)}$ then it reduces to $h^{(7)}$ so that the class $h^{(6)}$ includes $h^{(7)}$.

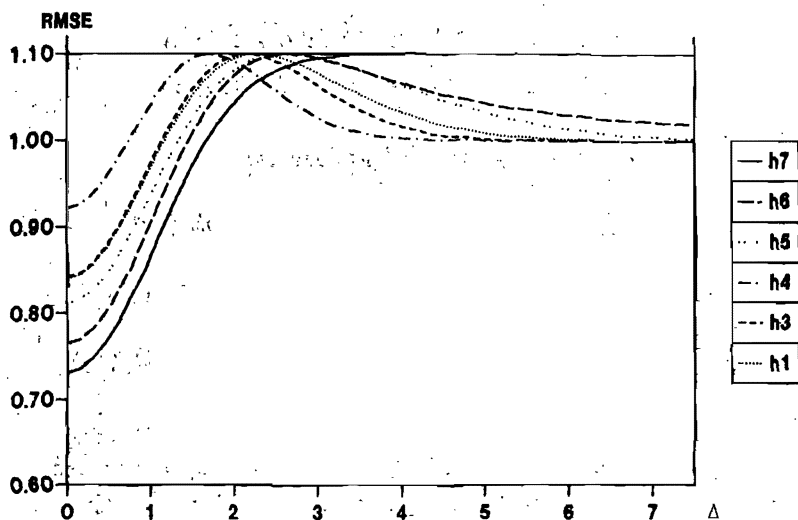
TABLE 1
RMSE AT 0 FOR GIVEN MAXIMAL OVERALL RMSE

h	Max = 1.05		Max = 1.10		Max = 1.20	
	c	RMSE(0)	c	RMSE(0)	c	RMSE(0)
$h^{(1)}$	0.1954	0.8999	0.3625	0.8416	0.6498	0.8003
$h^{(3)}$	-	-	0.0929	0.8429	0.8556	0.8342
$h^{(4)}$	0.7091	0.9591	0.9080	0.9218	1.1731	0.8556
$h^{(5)}$	0.2977	0.8519	0.4800	0.8123	1.2605	0.9375
$h^{(6)}$	3.0750	0.8553	1.3128	0.7656	0.5928	0.6718
$h^{(7)}$	-	0.7935	-	0.7314	-	0.6624

Table 1 was computed numerically using subroutines DUVMIF and DQDAG of IMSL. For the estimators corresponding to $h^{(1)}, h^{(3)}, \dots, h^{(7)}$, it shows the RMSE's at $\Delta = 0$ for maximal RMSE's of 1.05, 1.10 and 1.20, and the relevant choices of c . In the case of $h^{(2)}$ the minimal maximal RMSE when c varies is 1.291 so that $h^{(2)}$ cannot be accommodated in Table 1. For $h^{(3)}$ the minimal maximal RMSE is 1.09 so that Table 1 shows only the entries for maxima of 1.10 and 1.20 for $h^{(3)}$. For $h^{(6)}$ the choice $r(y) = cy^2$ was made. It is evident that $h^{(7)}$ outperforms the others with $h^{(6)}$ in second place. For larger ϵ , $h^{(6)}$ and $h^{(7)}$ are nearly equivalent. Figure 1 compares the RMSE's as a function of Δ for the estimators of Table 1 for the case $\epsilon = 0.1$ (the functions are all symmetric in Δ ; so only $\Delta \geq 0$ need be shown). This illustrates that the LT estimator has an increasing RMSE as $|\Delta|$ increases, taking its maximum value at ∞ . This is in contrast with the other estimators which all achieve their maxima at finite Δ and do not have such a small RMSE when $\Delta = 0$.

Fig. 1 COMPARISON OF RMSE FOR CASE EQUAL KNOWN
VARIANCES
(max rmse=1.1)

Fig.1 COMPARISON OF RMSE FOR CASE EQUAL KNOWN VARIANCES
(max rmse=1.1)



3. THE CASE σ_1^2/n_1 AND σ_2^2/n_2 ARBITRARY BUT KNOWN

In the case where σ_1^2/n_1 and σ_2^2/n_2 are not identical but are still known, the problem of estimating θ_ℓ is no longer invariant under permutation of X_1 and X_2 and we have to consider estimators of the form (1.2), where the adjustment function depends on ℓ . This is intuitively acceptable since if for example $\sigma_1^2/n_1 > \sigma_2^2/n_2$, then X_2 estimates θ_2 with greater accuracy than X_1 estimates θ_1 . Hence, if X_2 should happen to be larger than X_1 (i.e. $\ell = 2$) we would tend to make a smaller adjustment in order to estimate θ_ℓ than when the reverse holds.

For the remainder of this paper we assume that $\sigma_1^2/n_1 \geq \sigma_2^2/n_2$. The reverse case can be handled similarly. Put $\tau = (\sigma_2^2/n_2)/(\sigma_1^2/n_1) \leq 1$. The MSE of an estimator $X_\ell - h_\ell(|X_1 - X_2|)$ is given by

$$(3.1) E[X_1 - \theta_1 - h_1(vY)]^2(Y \geq 0) + E[X_2 - \theta_2 - h_2(-vY)]^2 I(Y < 0)$$

where

$$(3.2) \quad Y = (X_1 - X_2)/v \text{ and } v^2 = \sigma_1^2/n_1 + \sigma_2^2/n_2$$

so that Y is $N(\Delta, 1)$ -distributed, with $\Delta = (\theta_1 - \theta_2)/v$. As in the previous paragraph taking conditional expectations given Y , (3.1) becomes

$$(3.3) \quad \begin{aligned} & (\sigma_1^2/n_1)\{\tau + E[Y - \Delta - \tilde{h}_1(Y)]^2 I(Y \geq 0) \\ & + \tau^2 E[Y - \Delta + \tilde{h}_2(-Y)]^2 I(Y < 0)\}/(1 + \tau) \end{aligned}$$

where the two standardized adjustment functions are

$$(3.4) \quad \tilde{h}_1(y) = (vn_1/\sigma_1^2)h_1(vy) \text{ and } \tilde{h}_2(y) = (vn_2/\sigma_2^2)h_2(vy).$$

By taking $h_1(y) = h_2(y) = 0$, the MSE of X_L itself is found to be

$$(3.5) (\sigma_1^2/n_1)\{\tau + E[Y - \Delta]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta]^2 I(Y < 0)\}/(1 + \tau).$$

This MSE is strictly increasing as a function of Δ , tends to $\sigma_2^2/n_2 = \tau\sigma_1^2/n_1$ as $\Delta \rightarrow -\infty$ and to σ_1^2/n_1 as $\Delta \rightarrow \infty$. In the appendix we show that X_L is still minimax and admissible within the class of estimators defined by (1.2). Therefore the minimax value is still σ_1^2/n_1 and the RMSE of any estimator (1.2) is obtained by dividing (3.3) by σ_1^2/n_1 , i.e.

$$(3.6) \quad R(h_1, h_2, \Delta) = \begin{aligned} & \{\tau + E[Y - \Delta - \tilde{h}_1(Y)]^2 I(Y \geq 0) \\ & + \tau^2 E[Y - \Delta + \tilde{h}_2(-Y)]^2 I(Y < 0)\}/(1 + \tau) \end{aligned}$$

while the RMSE of X_L itself is

$$(3.7) \quad \begin{aligned} R(0, 0, \Delta) &= \{\tau + E[Y - \Delta]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta]^2 I(Y < 0)\}/(1 + \tau) \\ &= \tau + (1 - \tau)[\Phi(\Delta) - \Delta\phi(\Delta)] \end{aligned}$$

after some simplification. This function increases from τ at $\Delta = -\infty$ to 1 as $\Delta \rightarrow \infty$.

Admissibility of X_L implies that there is no estimator of the form (1.2) that has RMSE uniformly below (3.7). However the possibility still exists to improve on X_L when $\theta_1 = \theta_2$. As before the exact optimality problem is to

find adjustment functions h_1 and h_2 which minimize $R(h_1, h_2, 0)$ subject to the restriction that $R(h_1, h_2, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ for all Δ . When $\tau < 1$ this problem no longer reduces to Bickel's problem P and its exact solution is unknown. It seems difficult to apply the first two of Bickel's approximate optimality principles (Bickel (1984) p.868-869) but the third leads directly to an extended LT estimator for this case.

The third principle entails minimizing $R(h_1, h_2, 0)$ subject to bounded adjustment functions h_1 and h_2 . This is equivalent to minimizing

$$(3.8) \quad E[Z - \tilde{h}_1(Z)]^2 I(Z \geq 0) + \tau^2 E[Z + h_2(-Z)]^2 I(Z < 0)$$

with $ZN(0, 1)$ -distributed, subject to

$$(3.9) \quad \tilde{h}_1(y) \leq q_1 \text{ and } \tilde{h}_2(y) \leq q_2 \text{ for all } y \geq 0$$

with q_1 and q_2 two given positive bounds. Since (3.8) is

$$\begin{aligned} & E[Z - \tilde{h}_1(Z)]^2 I(0 \leq Z \leq q_1) + E[Z - \tilde{h}_1(Z)]^2 I(Z > q_1) + \\ & \tau^2 E[Z + \tilde{h}_2(-Z)]^2 I(-q_2 \leq Z \leq 0) + \tau^2 E[Z + \tilde{h}_2(-Z)]^2 I(Z < -q_2) \\ & \geq E[Z - q_1]^2 I(Z > q_1) + \tau^2 E[Z + q_2]^2 I(Z < -q_2) \end{aligned}$$

with equality holding for the Efron-Morris functions

$$(3.10) \quad \tilde{h}_i(y) = \tilde{h}_{q_i}(y) = yI(0 \leq y \leq q_i) + q_i I(y > q_i), \quad i = 1, 2,$$

the adjustment functions related to (3.10) by (3.4) yield the required approximately optimal solutions. They are $h_{q_i}(y) = (\sigma_i^2/vn_i)[(y/v)I(0 \leq y \leq vq_i) + q_i I(y > vq_i)]$, $i = 1, 2$ and the corresponding estimator becomes

$$(3.11) \quad X_t - h_{q_t}(|X_1 - X_2|) = X_t - (\sigma_t^2/n_t v) \tilde{h}_{q_t}(|X_1 - X_2|/v).$$

Its RMSE, $R(h_{q_1}, h_{q_2}, \Delta)$ is given by (3.6) with $\tilde{h}_1(Y)$ and $\tilde{h}_2(-Y)$ replaced by $\tilde{h}_{q_1}(Y)$ and $\tilde{h}_{q_2}(-Y)$ respectively. Interestingly enough, if we choose the bounds

$$(3.12) \quad q_1 = \sqrt{\epsilon(1 + \tau)} \text{ and } q_2 = \sqrt{\epsilon(1 + 1/\tau)}$$

then this RMSE again satisfies $R(h_{q_1}, h_{q_2}, \Delta) \leq (1+\epsilon)R(0, 0, \Delta)$ for all Δ , with equality when $|\Delta| \rightarrow \infty$. This is shown in part (iii) of the appendix. Consequently the third approximate optimality principle leads to a solution which does satisfy the restrictions of the exact optimality problem and we call the corresponding estimator the LT estimator for this case.

Estimators for this case can also be based on the families $h^{(j)}$ of §2. As an example take the Cohen-Sackrowitz family $h^{(6)}$. For the choice $r(y) = cy^2$ the standardized form of $h^{(6)}$ is given by (2.4) as

$$(3.13) \quad \bar{h}_c(y) = y/(1 + cy^2) \text{ for } y \geq 0.$$

Then two related adjustment functions are given by (3.4) as

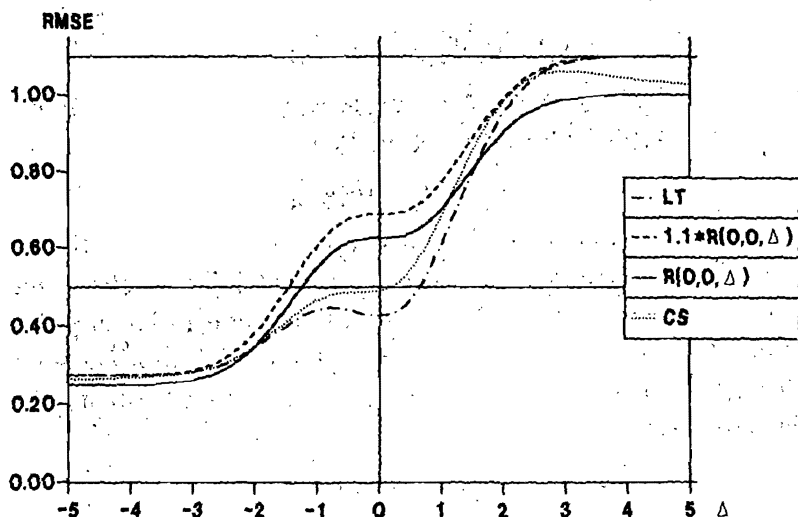
$$(3.14) \quad h_{c_1}(y) = (\sigma_1^2/n_1 v) \bar{h}_{c_1}(y/v) \text{ and } h_{c_2}(y) = (\sigma_2^2/n_2 v) \bar{h}_{c_2}(y/v).$$

The RMSE of the corresponding estimator $X_L - h_{c_1}(|X_1 - X_2|)$ is $R(h_{c_1}, h_{c_2}, \Delta)$ which can be found as in (3.6). As in the previous paragraph we now have to choose c_1 and c_2 so that $R(h_{c_1}, h_{c_2}, \Delta) \leq (1+\epsilon)R(0, 0, \Delta)$ for all Δ . This is a much more difficult numerical problem than before and in addition c_1 and c_2 depend on τ as well. In Figure 2 we show $R(h_{c_1}, h_{c_2}, \Delta)$ for the case $\tau = 0.25$ and $\epsilon = 0.1$ where we have chosen $c_1 = 3.0$ and $c_2 = 0.5$. The graph of $1.1R(0, 0, \Delta)$ is also shown and it is evident that this choice of c_1 and c_2 yields approximately $R(h_{c_1}, h_{c_2}, \Delta) \leq 1.1R(0, 0, \Delta)$ for all Δ . Also shown are $R(0, 0, \Delta)$, the RMSE of X_L , and $R(h_{q_1}, h_{q_2}, \Delta)$, the RMSE of the LT estimator. The improvement of the LT estimator on X_L for small Δ is clear. The Cohen-Sackrowitz estimator compares favorably with the LT estimator, but the difficulties in finding appropriate values for c_1 and c_2 compared with the ease with which q_1 and q_2 for the LT estimator are obtained, makes the Cohen-Sackrowitz estimator a less attractive option. Similar conclusions hold for other values of ϵ and τ .

Fig. 2 COMPARISON OF RMSE FOR CASE KNOWN BUT
ARBITRARY VARIANCES

($\tau = 0.25$, max rmse=1.1)

Fig.2 COMPARISON OF RMSE FOR CASE KNOWN BUT ARBITRARY VARIANCES
($\tau = 0.25$, max rmse=1.1)



4. THE CASE OF EQUAL UNKNOWN VARIANCES

In this paragraph we assume that $\sigma_1^2 = \sigma_2^2 = \sigma^2$ but this common value is unknown. Equivariance considerations suggest that we restrict attention to estimators of the form

$$(4.1) \quad X_t = Sh_t(|X_1 - X_2|/S)$$

where $S^2 = [(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]/m$, $m = n_1 + n_2 - 2$, is the usual unbiased estimator of σ^2 . For this case

$$(4.2) \quad v = \sigma\sqrt{1/n_1 + 1/n_2} = \sigma u \text{ and } \tau = n_1/n_2 \leq 1.$$

Put $T = S/\sigma$, so that MT^2 is χ_m^2 -distributed, independent of X_1, X_2 , and Y . Then the RMSE of (4.1) is

$$(4.3) \quad R(h_1, h_2, \Delta) = \frac{\{\tau + E[Y - \Delta - T\tilde{h}_1(Y/T)]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta + T\tilde{h}_2(-Y/T)]^2 I(Y < 0)\}}{(1 + \tau)}$$

with

$$(4.4) \quad \tilde{h}_i(y) = n_i u h_i(y/u) \quad i = 1, 2.$$

Again the exact optimization problem would be to find $\tilde{h}_1$ and $\tilde{h}_2$ which minimize $R(h_1, h_2, 0)$, subject to the restriction

$$R(h_1, h_2, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$$

for all Δ . This optimization problem appears formidable but since the LT estimators did very well in the previous cases, we try them again. This means that we take $\tilde{h}_i(y) = \tilde{h}_{q_i}(y)$ as in (3.10) with q_i as in (3.12) and solve for h_i from (4.4). Abusing notation we denote these adjustment functions again by $h_{q_i}(y)$. They are now given by $h_{q_i}(y) = (1/n_i u) \tilde{h}_{q_i}(y/u)$ and the resulting estimator is

$$X_t - (S/n_t u) \tilde{h}_{q_t}(|X_1 - X_2|/Su).$$

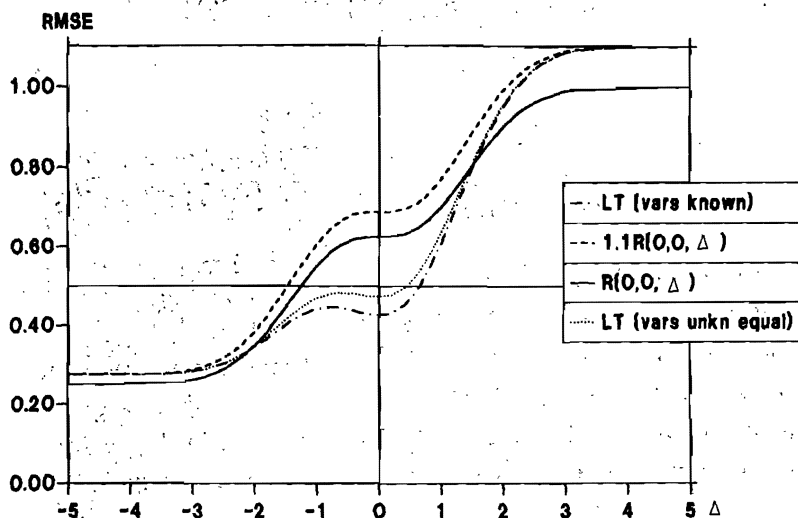
It is remarkable that the RMSE of this estimator still satisfies the restriction $R(h_{q_1}, h_{q_2}, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ for all Δ with equality when $|\Delta| \rightarrow \infty$. This is shown in part (iv) of the appendix.

It is easy to calculate $R(h_{q_1}, h_{q_2}, \Delta)$ as a numerical integral with respect to the densities of T and Y by using e.g. subroutine DQDAGI of IMSL. In Figure 3 we compare $R(h_{q_1}, h_{q_2}, \Delta)$ for the case $m = 1$ with the case where the variances are known. In both cases $\tau = 0.25$ and $\epsilon = 0.1$. This illustrates that m has very little influence on $R(h_{q_1}, h_{q_2}, \Delta)$. The same effect was found for other values of τ and ϵ . We conclude that the cost in terms of an increased RMSE of not knowing the common value of σ_1^2 and σ_2^2 is quite small and the LT estimator is still approximately optimal. Estimators based on e.g. the Cohen-Sackrowitz adjustment function can also be constructed and studied along the above lines. The problem of choosing the appropriate constants c_1 and c_2 would, however, be even more severe than before.

Fig. 3 COMPARISON OF RMSE FOR CASE UNKNOWN EQUAL AND KNOWN VARIANCES

($m = 1, \tau = 0.25, \max \text{rmse} = 1.1$)

Fig.3 COMPARISON OF RMSE FOR CASE UNKNOWN EQUAL AND KNOWN VARIANCES
($m=1, \tau=0.25, \max \text{rmse}=1.1$)



5. THE CASE OF ARBITRARY UNKNOWN VARIANCES

If σ_1^2 and σ_2^2 are arbitrary and unknown, equivariance considerations do not suggest a specific form for an estimator of θ_t . We therefore define the LT estimator by replacing σ_1^2 and σ_2^2 by their respective unbiased estimators, S_1^2 and S_2^2 , in (3.11), i.e.

$$(5.1) \quad X_t - (S_t^2/n_t V) \tilde{h}_{Q_t}(|X_1 - X_2|/V) = X_t - g_t(|X_1 - X_2|)$$

where

$$V^2 = S_1^2/n_1 + S_2^2/n_2, Q_1 = \sqrt{\epsilon(1 + (S_2^2/n_2)/(S_1^2/n_1))} \text{ and}$$

$$Q_2 = \sqrt{\epsilon(1 + (S_1^2/n_1)/(S_2^2/n_2))}$$

and $\tilde{h}_{Q_t}$ is given by (3.10) with q_i replaced by Q_t . Arguing as before the RMSE of (5.1) is given by

$$(5.2) \quad R(g_1, g_2, \Delta) = \frac{\{\tau + E[Y - \Delta - t_1^2 \tilde{h}_{U_1}(Y)]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta + t_2^2 \tilde{h}_{U_2}(-Y)]^2 I(Y < 0)\}}{(1 + \tau)}$$

where

$$\begin{aligned} T_1^2 &= S_1^2/\sigma_1^2, & T_2^2 &= S_2^2/\sigma_2^2 \\ t_1 &= T_1^2(1 + \tau)/(T_1^2 + \tau T_2^2), & t_2 &= T_2^2(1 + \tau)/(T_1^2 + \tau T_2^2), \\ U_1 &= \sqrt{\epsilon/(1 + \tau)}(T_1^2 + \tau T_2^2)/T_1, & U_2 &= \sqrt{\epsilon/\tau(1 + \tau)}(T_1^2 + \tau T_2^2)/T_2. \end{aligned}$$

It is possible to evaluate (5.2) as a trivariate integral with respect to the distributions of Y , T_1^2 and T_2^2 by using e.g. DQAND of IMSL.

It turns out that even for this case it is still true the RMSE of (5.1) as given by (5.2) never exceeds $(1 + \epsilon)R(0, 0, \Delta)$ for any Δ with equality being achieved as $|\Delta| \rightarrow \infty$. This is shown in part (v) of the appendix. Numerical evidence shows that $R(g_1, g_2, \Delta)$ differs very little from $R(h_{q_1}, h_{q_2}, \Delta)$, i.e. again the cost of not knowing σ_1^2/n_1 and σ_2^2/n_2 is small. As an example, Fig. 4 shows these RMSE's for the case $n_1 = n_2 = 2$, $\tau = 0.25$ and $\epsilon = 0.1$. For larger values of n_1 and n_2 the two RMSE's are even closer together. The approximate optimality of the LT estimator (3.11) for the case of known variances is therefore likely to hold for (5.1) as well.

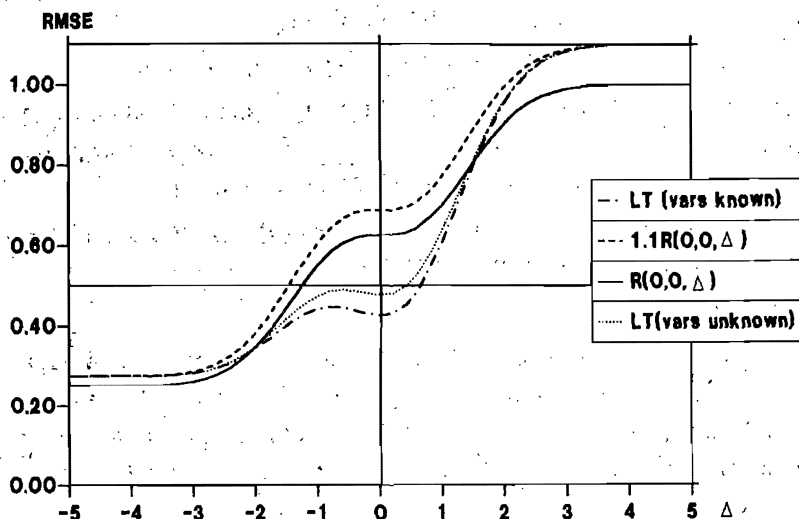
6. ACKNOWLEDGEMENT

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Fig. 4 COMPARISON OF RMSE FOR CASES OF UNKNOWN AND KNOWN VARIANCES

$(n_1 = 2, n_2 = 2, \tau = 0.25, \max \text{rmse} = 1.1)$

Fig.4 COMPARISON OF RMSE FOR CASES OF UNKNOWN AND KNOWN VARIANCES
 $(n_1 = 2, n_2 = 2, \tau = 0.25, \max \text{rmse} = 1.1)$



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APPENDIX

(i) **Minimaxity of X_L** : Consider the case of §3 with σ_1^2/n_1 and σ_2^2/n_2 arbitrary known values. Put an $N(\mu_n, \alpha_n^2)$ -prior on θ_1 and an $N(-\mu_n, \alpha_n^2)$ -prior on θ_2 , independent of θ_1 , where $\alpha_n \rightarrow \infty$ and $\mu_n/\alpha_n \rightarrow \infty$ as $n \rightarrow \infty$. Then the risk (w.r.t. squared error loss) of the Bayes estimator of θ_L is found to be

$$\alpha_n^2(\sigma_1^2/n_1)\Phi(\Delta)/(\sigma_1^2/n_1 + \alpha_n^2) + \alpha_n^2(\sigma_2^2/n_2)\Phi(-\Delta)/(\sigma_2^2/n_2 + \alpha_n^2).$$

The corresponding Bayes risk r_n is the expectation of this expression with respect to the $N(2\mu_n/v, 2\alpha_n^2/v^2)$ -prior of Δ . It is easily seen that $r_n \rightarrow \sigma_1^2/n_1$ as $n \rightarrow \infty$. Since σ_1^2/n_1 is the maximal risk of X_L for this case, it follows from Lehmann (1983) (Theorem 2.2, p. 256) that X_L is minimax. Although this theorem is formulated only for "fixed parameters" it only deals with risk functions and is readily seen to apply also to "random parameters" such as θ_L .

(ii) **Admissibility of X_L** : Continuing with the same case, the standard argument to prove admissibility of the single sample mean can be adapted to show admissibility of X_L within the class of estimators defined by (1.2). (See e.g. Ferguson (1967) p.141 or Lehmann (1983) p.265). We assume that there exists an estimator within the class (1.2) that dominates X_L

and then derive a contradiction using independent $N(0, \alpha_n^2)$ -priors for θ_1 and θ_2 with $\alpha_n^2 \rightarrow \infty$ as $n \rightarrow \infty$.

(iii) Proof of $R(h_{q_1}, h_{q_2}, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ (case of §3): By (3.6) the RMSE of (3.11) is

$$(8.1) \quad R(h_{q_1}, h_{q_2}, \Delta) = \frac{\{\tau + E[Y - \Delta - \tilde{h}_{q_1}(Y)]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta + \tilde{h}_{q_2}(-Y)]^2 I(Y < 0)\}}{(1 + \tau)}$$

and by (3.7) the RMSE of X_t is

$$(8.2) \quad R(0, 0, \Delta) = \frac{\{\tau + E[Y - \Delta]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta]^2 I(Y < 0)\}}{(1 + \tau)}.$$

Now it is easily seen that

$$(8.3) \quad \begin{aligned} (1 + \epsilon)R(0, 0, \Delta) &= \frac{\{\tau + [1 + \epsilon(1 + \tau)]E[Y - \Delta]^2 I(Y \geq 0) + \tau^2[1 + \epsilon(1 + 1/\tau)]E[Y - \Delta]^2 I(Y < 0)\}}{(1 + \tau)} \\ &= \frac{\{\tau + (1 + q_1^2)E[Y - \Delta]^2 I(Y \geq 0) + \tau^2(1 + q_2^2)E[Y - \Delta]^2 I(Y < 0)\}}{(1 + \tau)}. \end{aligned}$$

Hence $R(h_{q_1}, h_{q_2}, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ is equivalent to

$$(8.4) \quad \begin{aligned} &E[Y - \Delta - \tilde{h}_{q_1}(Y)]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta + \tilde{h}_{q_2}(-Y)]^2 I(Y < 0) \\ &\leq (1 + q_1^2)E[Y - \Delta]^2 I(Y \geq 0) + \tau^2(1 + q_2^2)E[Y - \Delta]^2 I(Y < 0) \end{aligned}$$

which is equivalent to

$$\begin{aligned} &-2E\tilde{h}_{q_1}(Y)(Y - \Delta)I(Y \geq 0) + E\tilde{h}_{q_1}(Y)^2 I(Y \geq 0) \\ &+ 2\tau^2 E\tilde{h}_{q_2}(-Y)(Y - \Delta)I(Y < 0) + \tau^2 E\tilde{h}_{q_2}(-Y)^2 I(Y < 0) \\ &\leq q_1^2 E[Y - \Delta]^2 I(Y \geq 0) + \tau^2 q_2^2 E[Y - \Delta]^2 I(Y < 0). \end{aligned}$$

Using (3.10) and partial integration, this requirement is equivalent to

$$(8.5) \quad \begin{aligned} &\int_0^{q_1} (q_1^2 - y^2 + 2)\phi(y - \Delta)dy + \tau^2 \int_0^{q_2} (q_2^2 - y^2 + 2)\phi(y + \Delta)dy \\ &\geq \Delta\phi(\Delta)(q_1^2 - \tau^2 q_2^2) \\ &= \Delta\phi(\Delta)\epsilon(1 - \tau^2). \end{aligned}$$

This obviously holds for $\Delta \leq 0$. For $\Delta > 0$, the left hand side is at least

$$\int_0^{q_1} (q_1^2 - y^2 + 2)\phi(y - \Delta)dy = \phi(\Delta) \int_0^{q_1} (q_1^2 - y^2 + 2)\exp(-\frac{1}{2}y^2 + \Delta y)dy.$$

Using $\exp(\Delta y) \geq \Delta y$ and integrating partially, this becomes

$$\Delta\phi(\Delta)q_1^2 = \Delta\phi(\Delta)\epsilon(1 + \tau) \geq \Delta\phi(\Delta)\epsilon(1 - \tau^2)$$

so that (8.5) holds and the result follows.

(iv) **Proof of $R(h_{q_1}, h_{q_2}, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ (case of §4):** From (3.10), $T\tilde{h}_{q_1}(Y/T) = \tilde{h}_{T_{q_1}}(Y)$, and hence $R(h_{q_1}, h_{q_2}, \Delta)$ is given by (4.3) with $T\tilde{h}_1(Y/T)$ and $T\tilde{h}_2(-Y/T)$ replaced by $\tilde{h}_{T_{q_1}}(Y)$ and $\tilde{h}_{T_{q_2}}(-Y)$ respectively. Also note that by (3.12), $ET^2q_1^2 = \epsilon(1 + \tau)$ and $ET^2q_2^2 = \epsilon(1 + 1/\tau)$. Using independence of T and Y in (8.3) we therefore have

$$\begin{aligned} & (1 + \epsilon)R(0, 0, \Delta) \\ &= E\{\tau + (1 + T^2q_1^2)[Y - \Delta]^2 I(Y \geq 0) \\ &+ \tau^2(1 + T^2q_2^2)[Y - \Delta]^2 I(Y < 0)\}/(1 + \tau) \end{aligned}$$

and as above $R(h_{q_1}, h_{q_2}, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ is equivalent to

$$\begin{aligned} & E\left\{\int_0^{Tq_1} (T^2q_1^2 - y^2 + 2)\phi(y - \Delta)dy \right. \\ &+ \left. \tau^2 \int_0^{Tq_2} (T^2q_2^2 - y^2 + 2)\phi(y + \Delta)dy\right\} \\ &\geq \Delta\phi(\Delta)E(T^2q_1^2 - \tau^2T^2q_2^2) \\ &= \Delta\phi(\Delta)\epsilon(1 - \tau^2). \end{aligned}$$

Again this obviously holds for $\Delta \leq 0$ while for $\Delta > 0$ the previous argument needs only slight adaptation.

(v) **Proof of $R(g_1, g_2, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ (case of §5):** The argument for this case is similar. Note that from (5.3) $Et_1^2U_1^2 = \epsilon(1 + \tau)$ and $Et_2^2U_2^2 = \epsilon(1 + 1/\tau)$ and again by independence of T_1, T_2 and Y we have

$$\begin{aligned} & (1 + \epsilon)R(0, 0, \Delta) \\ &= E\{\tau + (1 + t_1^2U_1^2)[Y - \Delta]^2 I(Y \geq 0) \\ &+ \tau^2(1 + t_2^2U_2^2)[Y - \Delta]^2 I(Y < 0)\}/(1 + \tau). \end{aligned}$$

Also

$$R(g_1, g_2, \Delta) = \{\tau + E[Y - \Delta - t_1^2 \tilde{h}_{U_1}(Y)]^2 I(Y \geq 0) + \tau^2 E[Y - \Delta + t_2^2 \tilde{h}_{U_2}(-Y)]^2 I(Y < 0)\} / (1 + \tau)$$

and $R(g_1, g_2, \Delta) \leq (1 + \epsilon)R(0, 0, \Delta)$ is equivalent to

$$\begin{aligned} (8.6) \quad E\{ & \int_0^{U_1} (t_1^2 U_1^2 - t_1^2 y^2 + 2t_1) \phi(y - \Delta) dy \\ & + \tau^2 \int_0^{U_1} (t_2^2 U_2^2 - t_2^2 y^2 + 2t_2) \phi(y + \Delta) dy \} \\ & \geq \Delta \phi(\Delta) E(t_1^2 U_1^2 - \tau^2 t_2^2 U_2^2) \\ & = \Delta \phi(\Delta) \epsilon (1 - \tau^2). \end{aligned}$$

Once more, for $\Delta \leq 0$ this obviously holds. For $\Delta > 0$, the left hand side is at least the first term, which equals

$$(8.7) \quad \phi(\Delta) E t_1^2 \int_0^{U_1} (U_1^2 - y^2 + 2/t_1) \exp(-\frac{1}{2}y^2 + \Delta y) dy.$$

From (5.3) $t_1 \leq 2 < e$ and using also the inequality $\exp(\Delta y) \geq e\Delta y$, the integral in (8.7) is at least

$$\begin{aligned} & e\Delta \int_0^{U_1} (U_1^2 - y^2 + 2/e)y \exp(-\frac{1}{2}y^2) dy \\ & = e\Delta [U_1^2 + 2(1 - 1/e)(\exp(-\frac{1}{2}U_1^2) - 1)] \\ & \geq e\Delta [U_1^2 + 2(1 - 1/e)(-\frac{1}{2}U_1^2)] \\ & = \Delta U_1^2. \end{aligned}$$

Hence (8.7) is at least $\Delta \phi(\Delta) E t_1^2 U_1^2 = \Delta \phi(\Delta) \epsilon (1 + \tau) \geq \epsilon \Delta \phi(\Delta) (1 - \tau^2)$ and (8.6) holds again.